

The Autocorrelation Mirage: Dependence-Aware Evaluation for Panel-Expanded Event Forecasting

Abstract. Panel expansion can turn a small collection of independent episodes into many dependent rows. When row-wise cross-validation places rows from an episode in both training and test folds, it answers an interpolation question even when deployment requires generalization to unseen episodes. We call the resulting mechanism *episode memorization*; latent episode heterogeneity and shared event-time structure, not autocorrelation alone, drive the effect. In a controlled simulation with strictly causal features, row-wise five-fold cross-validation raises boosted-tree AUROC from 0.552 to 0.835 ($\Delta_{CV} = 0.283$, paired 95% CI [0.246, 0.320]). A higher-signal configuration has nontrivial grouped AUROC (0.723), yet retains a 0.199 gap (95% CI [0.170, 0.229]). A 10-replicate-per-cell grid finds larger gaps for tree ensembles than logistic regression. A separate drift-based DGP (30 replicates per model) finds a smaller normalization-leakage effect; it is distinguishable from zero only for random forests. We propose reporting grouped and row-wise estimates, fold-local preprocessing, and Δ_{CV} as a practical diagnostic. All empirical evidence is simulation-based.

Keywords: pseudoreplication, cross-validation, panel data, event forecasting, episode memorization

1 Introduction

Consider a hospital forecasting which admitted patients risk deterioration within two weeks. Clinicians have followed 30 patients closely enough to have detailed daily vitals – only 30 independent trajectories, far too few to train a model with one row per patient. A natural workaround expands each stay into one row per day, turning 30 episodes into hundreds of rows and making the pipeline look like an ordinary tabular-classification problem. The same workaround recurs in predictive maintenance, expanding a handful of machine failure histories into daily sensor rows, and in conflict early-warning systems, expanding a few dozen countries into a denser country-month panel [2, 6]. This workaround – panel expansion under a small number of truly independent units – motivates the rest of the paper: does the resulting row count reflect real independent evidence, and does standard cross-validation treat it that way?

Trustworthy AI systems depend not only on model capacity but also on evaluation protocols that match the deployment target. In each of the settings above, the analyst typically has only 20–50 independent episodes and 10–20 features,

already a small-data regime by conventional events-per-variable heuristics for generalized linear models [8, 12], and the expansion into rows does not fix that.

That expansion does not simply increase sample size; it changes the evaluation problem. When row-wise K-Fold cross-validation assigns expanded observations to folds independently, rows from the same episode can appear in both training and test sets. Because consecutive observations $X_{e,t}$ and $X_{e,t+1}$ are autocorrelated, a flexible model can recognize episode-specific trajectory structure rather than infer risk that transfers to unseen episodes. We call this mechanism **episode memorization**. In that setting, row-wise CV turns a forecasting task (generalizing to new episodes) into a partially interpolative task (matching rows within known episodes).

It is useful to distinguish three failure modes that are often conflated. *Explicit leakage* occurs when a feature directly encodes unavailable future information such as $T_e - t$. *Normalization leakage* occurs when test-fold statistics or future within-episode summaries enter preprocessing even though they would be unavailable at prediction time. *Pseudoreplication* occurs when dependent rows from the same episode are split across training and test folds. The first two are feature- or preprocessing-level violations. The third is an evaluation-design violation and can arise even when every feature is strictly causal.

In our simulation, replacing grouped five-fold evaluation with row-wise KFold increases mean AUC by 0.283, from 0.552 to 0.835. In the same benchmark, an explicit time-to-event leak increases grouped-evaluation AUC by 0.261, from 0.552 to 0.813. We present this comparison as evidence that pseudoreplication can be severe, not as a universal ranking across all domains or models. This finding builds on established work on data leakage, pseudoreplication, and blocked or subject-wise cross-validation, which we review in Section 2 together with our contribution relative to it.

The paper makes six contributions. First, it clarifies the distinction between explicit leakage, preprocessing leakage, and pseudoreplication in expanded panels. Second, it formalizes episode memorization as the mechanism by which row-wise CV can produce optimistic estimates. Third, it quantifies the resulting optimism in a controlled benchmark with strictly causal features, and shows that the gap persists even in a higher-signal configuration where grouped-CV performance is no longer near chance. Fourth, it stress-tests that optimism over a compact grid of episode counts, dependence levels, dimensionalities, and model families. Fifth, it demonstrates an analogous optimism from preprocessing leakage under a second, drift-based data-generating process. Sixth, it proposes Δ_{CV} , the gap between row-wise and grouped CV estimates, as a practical warning signal together with a dependence-aware evaluation protocol and an audit checklist.

The goal is methodological rather than epidemiological: we identify a failure mode, explain its mechanism, and quantify its potential impact under controlled conditions. We do not attempt to measure how prevalent this problem is in published practice.

The rest of the paper proceeds as follows: related work (Sec. 2); the theoretical framework and three failure modes (Sec. 3); implicit normalization and

imputation traps (Sec. 4); the episode-memorization mechanism and a sibling-overlap result (Sec. 5); the empirical benchmark, higher-signal check, robustness grid, and drift experiment (Sec. 6); the Δ_{CV} diagnostic and evaluation protocol (Sec. 7); scope, limitations, and the audit checklist (Sec. 8); and conclusions (Sec. 9).

2 Related Work

Leakage taxonomies. Kaufman et al. [5] formalized data leakage in i.i.d. settings through a learn/predict separation criterion; Section 3 extends that criterion to filtrations and adds pseudoreplication as a third failure mode distinct from the explicit and preprocessing leakage that Kapoor and Narayanan [4] found dominate published leakage audits.

Blocked, grouped, and subject-wise cross-validation. Roberts et al. [10] argued for block cross-validation under temporal, spatial, hierarchical, or phylogenetic structure, and de Prado [9] proposed purged and embargoed cross-validation for overlapping labels in finance, a related but distinct problem (label overlap rather than expanded-row dependence). Hurlbert [3] coined *pseudoreplication* for non-independent replicates analyzed as if independent, the direct ancestor of our use of the term. Closer to our setting, Saeb et al. [11] and Vabalas et al. [13] showed empirically that record-wise cross-validation overstates performance relative to subject-wise validation in clinical and wearable-sensor ML. These studies document the phenomenon within one application each; we instead isolate the mechanism analytically (Section 5) and quantify it against a matched explicit-leakage benchmark in event-time panel data.

Panel-expanded conflict forecasting. Muchlinski et al. [6] and the ViEWS system [2] popularized flexible ML on panel-expanded conflict data, and Cranmer and Desmarais [1] and Neunhoffer and Sternberg [7] showed that cross-validation misuse can overturn substantive claims in that literature. We share their motivation but contribute a domain-agnostic mechanism and diagnostic rather than a re-analysis of one dataset.

Relative to all of the above, no prior study isolates pseudoreplication’s effect size from explicit and preprocessing leakage under one shared, controlled protocol stress-tested across dependence, episode counts, and model families (Section 9).

3 Theoretical Framework

Consider E episodes indexed by e . Each has time $t \in \{0, \dots, T_{\max}\}$, administrative censoring time C_e , observed-event indicator $\delta_e := \mathbf{1}\{\text{event observed by } C_e\}$, features $X_{e,t} \in \mathbb{R}^p$, and static attributes α_e . When $\delta_e = 1$, let T_e denote the observed event time. For horizon H , eligible rows are

$$\mathcal{T}_e(H) := \{t : \delta_e = 1, t < T_e\} \cup \{t : \delta_e = 0, t + H \leq C_e\}.$$

The binary target is

$$Y_{e,t}(H) := \begin{cases} \mathbf{1}\{T_e \leq t + H\}, & \delta_e = 1, \\ 0, & \delta_e = 0 \text{ and } t + H \leq C_e. \end{cases}$$

Late censored rows with $t + H > C_e$ are excluded from the supervised evaluation set rather than labeled heuristically.

Let $\mathcal{F}_{e,t-} := \sigma(\{X_{e,s} : s < t\} \cup \{\alpha_e\} \cup \mathcal{E}_{<t})$ denote information strictly before t .

Definition 1 (Predictable Feature). *Feature $X_{e,t}$ is predictable if it is $\mathcal{F}_{e,t-}$ -measurable, that is, determined by information available before t .*

This extends Kaufman et al.'s [5] learn-predict separation to filtrations. Violations need not be explicit: global statistics computed over $\{X_{e,s} : s \leq T_e\}$ implicitly depend on T_e .

Definition 2 (Explicit Leakage). *An observation exhibits explicit leakage if a feature is not predictable because it is a direct function of future outcome information such as $T_e - t$.*

Definition 3 (Normalization Leakage). *Normalization leakage occurs when test-fold statistics, future within-episode values, or full-trajectory episode summaries enter transformed inputs even though they would be unavailable at prediction time.*

Definition 4 (Pseudoreplication). *Pseudoreplication occurs when the evaluation split places observations from the same episode in both training and test folds, violating the independence assumptions underlying row-wise validation even if all features are predictable.*

The deployment target for unseen episodes is the episode-level risk

$$R_{\text{episode}}(f) = \mathbb{E}_{e^*} \left[\frac{1}{|\mathcal{T}_{e^*}|} \sum_{t \in \mathcal{T}_{e^*}} \ell\{f(X_{e^*,t}), Y_{e^*,t}\} \right],$$

where e^* is not represented in the training episodes. Episode-grouped validation estimates this target by holding out complete episodes. Row-wise validation instead estimates a conditional interpolation risk closer to

$$R_{\text{row}}(f) = \mathbb{E}[\ell\{f(X_{e,t}), Y_{e,t}\} \mid e \in \mathcal{E}_{\text{train}}],$$

because test rows may share latent episode effects, trajectory structure, and event time with training rows. Thus the issue is not dependence alone; it is a mismatch between the generalization unit implied by deployment and the unit used by validation.

Table 1. Three distinct sources of evaluation optimism in expanded panel forecasting.

Failure mode	Mechanism	Future info?	Primary fix
Explicit leakage	Feature directly reveals a post- t outcome signal such as $T_e - t$	Yes	Redesign or remove feature
Normalization leakage	Test-fold statistics, future within-episode values, or full-trajectory episode summaries enter transformed inputs	Indirectly	Fit transforms within each training fold; for online use, rely on trailing or historically available statistics
Pseudoreplication	Rows from the same episode appear in both train and test folds	No	Group by episode at split time

Remark 1 (Explicit leakage can make ranking nearly perfect). Suppose both classes have positive probability and the leaked score has the form $S_{e,t} = f(T_e - t) + \epsilon_{e,t}$, where f is strictly monotone on the support relevant for $Y_{e,t}(H)$ and the learned scoring rule uses the sign convention that smaller time-to-event implies higher risk. Restricting attention to eligible rows from episodes with observed event times (censored no-event rows can be treated as having $T_e - t = \infty$), $Y_{e,t}(H) = 1$ iff $T_e - t \in (0, H]$, so a strictly monotone transform of $T_e - t$ induces the same ordering as the raw time-to-event. If $\text{Var}(\epsilon_{e,t}) = \sigma^2$, then as $\sigma^2 \rightarrow 0$ the leaked score preserves that ordering almost surely and $\text{AUC} \rightarrow 1$. We record this as a sanity check rather than a substantive result: it confirms that a feature built directly from the outcome-defining quantity $T_e - t$ is exactly the kind of explicit leakage that Definition 2 rules out, and it calibrates the explicit-leak condition in Table 2 as a near-worst-case comparator rather than a typical one.

4 Implicit Violations in Standard Pipelines

Table 1 summarizes the three failure modes emphasized in this paper. Explicit leakage and normalization leakage are violations of predictability. Pseudoreplication is different: it is an evaluation mismatch induced by splitting dependent rows across folds.

Normalization traps are heterogeneous. Simple global standardization $\tilde{X} = (X - \mu)/\sigma$ can be weak when the feature process is approximately stationary, but transforms that use future within-episode values—for example, full-trajectory episode means or episode-wise z-scores—can be much more problematic. Mean imputation uses future observations; episode-mean imputation uses $\bar{X}_e$, which also depends on T_e . Centered moving averages use future values by construction. These traps require features with pre-existing trends toward the event. Strictly causal stationary features are often less sensitive, but many real-world variables

are not: vital signs may deteriorate before death, equipment diagnostics may worsen before failure, and conflict indicators may escalate before violence.

5 Episode Memorization Under Row-Wise Cross-Validation

Even with leak-free features, standard K-Fold CV can inflate performance through *episode memorization* when the target is generalization to unseen episodes. This is Hurlbert’s [3] pseudoreplication translated to ML: “replicates are not statistically independent.” Within-episode observations share: (1) latent episode effects α_e ; (2) autocorrelated features with $\text{Corr}(X_{e,t}, X_{e,t+1}) \gg 0$; (3) labels determined by the same T_e . When row-wise CV places observations from episode e in both training and test sets, the model learns episode-specific patterns—the “texture” of each episode’s autocorrelation structure. At test time, it performs *identity inference*: recognizing which episode an observation belongs to, then recalling the associated outcome. This transforms forecasting (extrapolation) into interpolation (pattern-matching). As a descriptive design-effect approximation, if each episode contributes approximately m rows and the average within-episode pairwise correlation is $\bar{\rho}$, then

$$n_{\text{eff}} \approx \frac{n}{1 + (m - 1)\bar{\rho}}.$$

This approximation is not used for inference; it is reported only to highlight that panel expansion can substantially overstate the amount of independent evidence.

A minimal example makes the mechanism concrete. Suppose episodes A, B, and C each contribute two daily rows. A random two-fold split may train on $\{A_1, B_1, C_2\}$ and test on $\{A_2, B_2, C_1\}$; every test row now has a sibling row from the same episode in training. A grouped split instead trains on $\{A_1, A_2, B_1, B_2\}$ and tests on $\{C_1, C_2\}$, forcing prediction to transfer across episodes rather than match within them.

For $n = 600$, $m \approx 20$, and $\bar{\rho} \approx 0.5$, this approximation gives $n_{\text{eff}} \approx 55$ —a $10\times$ reduction. But the inflation from episode memorization is *not* merely a sample-size effect: in the empirical benchmark below, row-wise CV still achieves AUC above 0.8 despite the small effective sample size.

This suggests an empirical expectation that, once within-episode dependence is present, row-wise CV will exceed grouped CV more strongly for higher-capacity models. Section 6 tests that expectation directly over a compact grid of dependence levels, episode counts, dimensionalities, and model families.

Proposition 1 (Sibling overlap under shuffled row-wise K-fold). *Consider an episode with m_e eligible rows assigned independently to one of K folds. Condition on one row being placed in the test fold. Then the probability that at least one of its $m_e - 1$ sibling rows appears in the training folds is*

$$\Pr(\text{sibling overlap}) = 1 - K^{-(m_e-1)},$$

and the expected number of same-episode training siblings is

$$\mathbb{E}[N_{\text{sibling in train}}] = (m_e - 1) \frac{K - 1}{K}.$$

Proof sketch. Condition on the test row being assigned to fold k . Its siblings avoid the training set only if they also fall in fold k , which under independent fold assignment occurs with probability $K^{-(m_e-1)}$. The complement gives the overlap probability, and linearity of expectation yields the expected sibling count. The exact probability differs slightly for deterministic balanced K-Fold, but the approximation captures the usual shuffled row-wise setting. $\square$

6 Empirical Validation

We generate $E = 30$ episodes with: episode effect $\alpha_e \sim \mathcal{N}(0, 0.5)$; AR(1) latent process $Z_{e,t} = 0.7Z_{e,t-1} + \epsilon_t$; hazard $h_{e,t} = \text{logit}^{-1}(-3 + \alpha_e + 0.15Z_{e,t})$; features from $Z_{e,<t}$ only (strictly causal); and administrative censoring $C_e = T_{\max} = 60$. For observed-event episodes ($\delta_e = 1$), eligible rows satisfy $t < T_e$ and receive label $Y_{e,t}(14) = \mathbf{1}\{T_e \leq t + 14\}$. For censored no-event episodes ($\delta_e = 0$), only rows with $t + 14 \leq C_e$ are eligible, and those rows receive $Y_{e,t}(14) = 0$; late censored rows are excluded rather than encoded through a synthetic $T_e = C_e + 1$. The intercept -3 corresponds to a low instantaneous daily hazard ($\text{logit}^{-1}(-3) \approx 0.047$), but the 14-day horizon makes the induced row-level label much denser: over 10 seeds, the filtered at-risk panel contains about 587 observations, mean episode length 19.6, episode-event frequency 0.91, and positive-row prevalence 0.48. We therefore frame the paper as panel-expanded event forecasting rather than as a benchmark for ultra-rare row labels. The five base features are strictly causal transformations of $Z_{e,<t}$; for the robustness grid we add independent nuisance features to reach $p \in \{20, 100\}$. The main benchmark uses a histogram gradient boosting classifier with 100 boosting iterations, maximum depth 4, and learning rate 0.1. Grouped and row-wise protocols both use $K = 5$ for the main benchmark: `GroupKFold(n_splits=5)` and `shuffled KFold(n_splits=5)`. The repository also implements a temporal-blocked within-episode splitter for the alternative target of forecasting later rows of already observed episodes; we report that baseline below in text because it estimates a different target than unseen-episode generalization. The compact robustness grid below uses grouped and row-wise three-fold CV for tractability. For each replicate, we compute out-of-fold predictions for all eligible rows and then compute a single pooled AUROC and Brier score. To keep longer episodes from dominating diagnostics, we also compute episode-weighted AUROC and episode-mean Brier summaries. This pooled evaluation avoids dropping grouped folds with one class, which can occur under episode-aware validation. Standardization is fit within each training fold except in the deliberate preprocessing-leakage conditions. Because each random seed is reused across grouped, row-wise, normalization, explicit-leak, and fold-prevalence conditions, the reported AUC gaps are paired within replicate. The reported 95% paired intervals use the matched replicate-level differences,

Table 2. Simulation results over 30 paired replicates. Also shown is a grouped training-fold prevalence predictor; the temporal-blocked within-episode baseline is reported in text because it targets a different estimand.

Condition	AUC	Brier	Δ AUC vs. baseline	95% paired CI
Leak-Free + Grouped CV	0.552 ± 0.101	0.330 ± 0.048	baseline	—
Training-Fold Prev. +	0.430 ± 0.025	0.245 ± 0.009	-0.122	[-0.159, -0.086]
Grouped CV				
Leak-Free + Row-wise KFold	0.835 ± 0.040	0.162 ± 0.023	+0.283	[0.246, 0.320]
Global Norm. + Grouped CV	0.552 ± 0.101	0.330 ± 0.048	+0.000 [†]	[0.000, 0.000]
Explicit Leak + Grouped CV	0.813 ± 0.042	0.198 ± 0.041	+0.261	[0.236, 0.286]

[†]See text for interpretation.

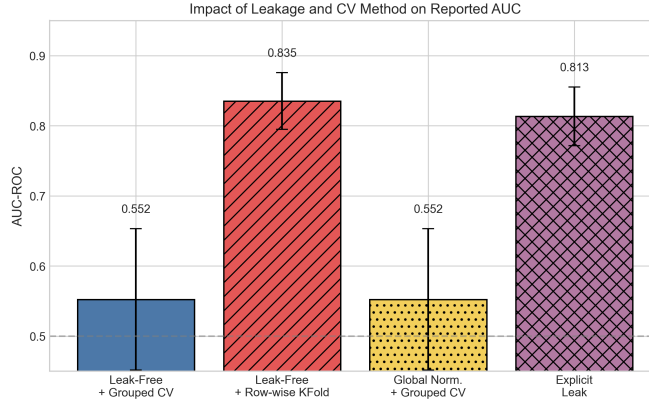


Fig. 1. Main benchmark (30 replicates) under matched five-fold protocols. Row-wise KFold increases AUROC and improves Brier score relative to episode-grouped five-fold validation even though the features are strictly causal.

$\bar{d} \pm t_{0.975, n-1} s_d / \sqrt{n}$. Because rows within episodes are dependent, we also report episode-bootstrap uncertainty for Δ_{CV} by resampling episode IDs rather than rows. All conditions are paired by random seed, and the same simulated episodes are used across grouped, fold-prevalence, row-wise, normalization, and explicit-leak settings within each replicate. An anonymized reproducibility package accompanies the submission, and the code, result CSVs, and figure-generation scripts will be released in a public repository upon acceptance. The main benchmark table, the higher-signal check (Section 6.1), and the drift-DGP normalization experiment each use 30 replicates; the robustness grid uses 10 replicates per cell. All replicate-level comparisons are paired by seed.

Table 2 and Figure 1 reveal the central finding: row-wise cross-validation alone can create substantial optimism. Row-wise KFold yields AUC=0.835 from strictly causal features, an absolute gap of 0.283 over grouped CV (AUC=0.552), with a matched-replicate 95% interval of [0.246, 0.320]. Episode-weighted sum-

Table 3. Matched-replicate CI for mean Δ_{CV} , and mean within-replicate episode-bootstrap interval in the 30-replicate main benchmark.

Summary	Δ_{CV}	95% interval
Matched replicate mean	0.283	[0.246, 0.320]
Mean within-replicate bootstrap	0.283	[0.141, 0.413]

Table 4. Robustness grid: mean Δ_{CV} averaged over $E \in \{20, 50, 100\}$ and $p \in \{5, 20, 100\}$ (10 replicates per cell).

Model family	$\rho = 0.0$	$\rho = 0.6$	$\rho = 0.9$
Logistic regression	0.021	0.025	0.042
Random forest	0.125	0.144	0.175
Boosted trees	0.177	0.206	0.215

maries point the same way: grouped CV yields episode-weighted AUC=0.547 and episode-mean Brier=0.280, whereas row-wise KFold reaches 0.823 and 0.159. The mean within-replicate episode-bootstrap interval for Δ_{CV} is [0.141, 0.413], the median is [0.123, 0.386], and the smallest lower endpoint across replicates is 0.066, so all 30 replicate-level intervals remain positive (Table 3). In this benchmark, explicit $T_e - t$ leakage raises grouped-evaluation AUC by 0.261 with paired 95% interval [0.236, 0.286]. The training-fold prevalence predictor attains AUC=0.430 and Brier=0.245; because each grouped split predicts the training-fold prevalence for its held-out episodes, its scores vary across folds and need not yield AUC=0.5. Relative to that baseline, grouped boosting improves ranking but worsens Brier (0.330), indicating that grouped CV is near chance in discrimination and poorly calibrated in this small- E setting, whereas row-wise CV hides both problems. For the alternative target of forecasting future rows of already observed episodes, temporal-blocked within-episode validation yields AUC=0.917 and Brier=0.232; because it carries same-episode history into training, it is appropriate for future rows of known episodes rather than unseen-episode generalization. Simple global normalization shows no measurable AUC gap in this stationary DGP, so that row should be read as a conservative baseline rather than as evidence that preprocessing leakage is unimportant. For tree-based learners, simple affine rescaling is also expected to have limited effect; the drift-DGP experiment below provides the positive preprocessing example. The same benchmark still averages only about 68 effective observations despite roughly 587 at-risk rows, underscoring that row expansion does not create independent evidence.

6.1 Does the Gap Survive a Genuinely Non-Trivial Grouped Baseline?

The grouped baseline in Table 2 is itself near chance (AUC 0.552), leaving open whether the row-wise gap reflects a genuine evaluation artifact or simply that

Table 5. Δ_{CV} by episode count E instead of dependence, averaged over ρ and p (10 replicates per cell).

Model family	$E = 20$	$E = 50$	$E = 100$
Logistic regression	0.047	0.029	0.012
Random forest	0.177	0.150	0.117
Boosted trees	0.203	0.205	0.190

Table 6. Raising the episode-effect standard deviation from 0.5 to 1.5 (30 paired replicates each, DGP otherwise identical) more than doubles grouped-CV AUC while the row-wise gap remains large.

Configuration	Grouped AUC	Row-wise AUC	Δ_{CV} (95% CI)
Baseline (episode-effect sd = 0.5)	0.552 ± 0.101	0.835 ± 0.040	0.283 [0.246, 0.320]
Higher signal (episode-effect sd = 1.5)	0.723 ± 0.099	0.922 ± 0.039	0.199 [0.170, 0.229]

grouped CV has little real signal to lose. We address this with a second configuration that raises the standard deviation of the episode random effect α_e from 0.5 to 1.5, holding every other DGP parameter fixed. This targets *transferable* signal specifically: α_e sets each episode’s overall risk level and is observed, with small noise, through the static feature X_5 , so a model can learn the α_e –outcome relationship on training episodes and apply it to a held-out episode. We first tried the more direct route of increasing the coefficient β linking the AR(1) latent process $Z_{e,t}$ to the hazard; over 10 replicates per setting, grouped-CV AUC stayed near chance (0.537–0.580) across $\beta \in \{0.15, \dots, 1.3\}$, an almost ninefold range, because $Z_{e,t}$ is an episode-specific trajectory that does not transfer across episodes – amplifying β mostly inflates the row-wise memorization gap rather than the transferable signal, which itself explains why the original grouped baseline is weak. Raising episode-effect heterogeneity instead worked as intended: over 30 paired replicates, grouped-CV AUC reaches 0.723 ± 0.099 (event rate 0.361), a genuinely non-trivial level of discrimination, while row-wise KFold reaches 0.922 ± 0.039 . The gap remains large and clearly significant, $\Delta_{CV} = 0.199$, 95% interval [0.170, 0.229] (Table 6) – row-wise optimism is not an artifact of a deliberately weak-signal benchmark, though it shrinks once grouped CV has more real signal to work with.

Table 4 tests whether the main result is tied to one model or one dimensionality setting, now with ten replicates per cell rather than two. Across $81 E \times \rho \times p \times$ model cells, higher-capacity tree ensembles exhibit much larger row-wise optimism than logistic regression. Averaged over E and p , mean Δ_{CV} rises from 0.125 to 0.175 for random forests and from 0.177 to 0.215 for boosted trees as dependence strengthens, while logistic regression stays close to zero (0.021–0.042). Across the 81 cell-level means, logistic-regression gaps range from 0.004 to 0.079 (sd 0.021), random-forest gaps from 0.053 to 0.363 (sd 0.076), and boosted-tree gaps from 0.132 to 0.276 (sd 0.047) – markedly tighter than the

earlier two-replicate version of this grid, though the ordering across model families is unchanged. Varying p up to 100 nuisance features does not remove the effect. Table 5 reports the same gap by episode count instead of dependence: mean Δ_{CV} falls somewhat as E grows from 20 to 100 for every model family, consistent with the grouped-CV estimate becoming more stable as more episodes are available, though the tree-ensemble gap remains large even at $E = 100$.

The nonzero gap at $\rho = 0$ shows that the effect is not driven by autocorrelation alone. Episode-level latent heterogeneity and shared event-time structure can already create row-wise optimism; increasing autocorrelation strengthens the same evaluation-unit mismatch.

Normalization leakage was tested under a second DGP with a monotone pre-event drift term: $Z_{e,t}$ is replaced by $Z_{e,t} + \text{drift_strength} \cdot (t/T_{\max})$, $\text{drift_strength} = 1.0$, so every episode’s expected latent level rises linearly over its observed span; all other parameters ($E = 30$, AR coefficient 0.7, hazard_coef 0.15, episode-effect sd 0.5, horizon 14) match the main benchmark. Over 30 replicates per model family, paired by seed, full-trajectory episode-wise normalization raises grouped-CV AUC relative to leak-free inputs by +0.045 for logistic regression (95% CI $[-0.010, 0.100]$), +0.063 for random forests (95% CI $[0.029, 0.096]$), and +0.034 for boosted trees (95% CI $[-0.010, 0.077]$); the random-forest shift is clearly distinguishable from zero, and the other two are borderline, with lower confidence bounds a few thousandths below zero. Global standardization had no measurable effect for any model, as expected for an affine transform. Earlier lightly replicated pilot estimates were considerably noisier and are not reported. Pre-processing leakage here is real, small, and consistently signed, and estimating its size required substantially more replicates than the main pseudoreplication result – the same caution we ask readers to apply to any single-digit-replicate Δ_{CV} estimate.

Figure 2 complements Table 4 by making the two supporting patterns visually explicit: larger Δ_{CV} values are concentrated in tree-based learners once dependence is moderate or high, and in the drift DGP grouped-CV AUC under full-trajectory episode normalization is higher than under leak-free inputs for every model family, most clearly for random forests.

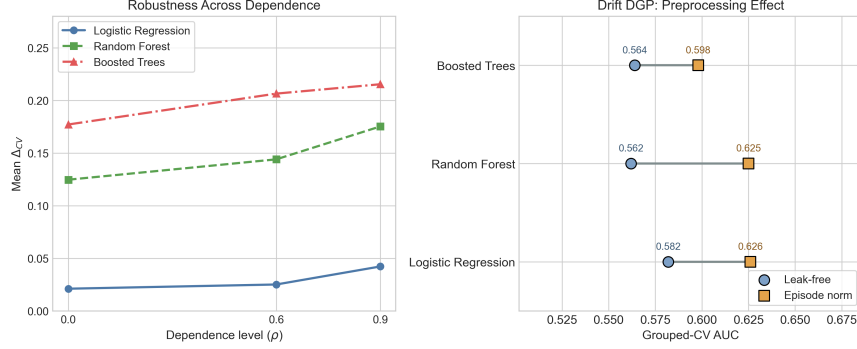


Fig. 2. Companion diagnostics for the strengthening experiments. Left: mean Δ_{CV} versus dependence after averaging over E and p for each model family. Right: grouped-CV AUC in the drift DGP under leak-free inputs and full-trajectory episode normalization; global standardization overlaps the leak-free baseline and is omitted for clarity.

7 The Δ_{CV} Diagnostic and Dependence-Aware Evaluation Protocol

We propose a simple diagnostic for split sensitivity under panel expansion:

Definition 5 (Δ_{CV} diagnostic). Let $\widehat{AUC}_{row}$ denote the pooled out-of-fold AUROC under row-wise KFold and let $\widehat{AUC}_{group}$ denote the pooled out-of-fold AUROC under episode-grouped KFold. We define

$$\Delta_{CV} = \widehat{AUC}_{row} - \widehat{AUC}_{group}.$$

Large positive values indicate split sensitivity: the model performs better when test rows are allowed to share episodes with training rows than when entire episodes are held out.

In our main benchmark, $\Delta_{CV} = 0.283$. In the robustness grid, logistic regression usually stays near zero, whereas random forests and boosted trees remain well above 0.10 on average and grow with ρ . That pattern makes values around 0.05 a useful warning heuristic in our setting, but we do not claim a universal decision threshold. The relevant magnitude will depend on the domain, target definition, dependence strength, and model family. This metric requires no additional data collection—only re-running evaluation with episode-grouped validation instead of row-wise KFold.

Dependence-Aware Evaluation Protocol: (1) *Preprocessing:* fit transformations on training folds only; use forward-fill imputation and trailing windows. (2) *Evaluation:* use grouped splitting with `groups=episode_id`; never split episodes across folds. (3) *Reporting:* report Δ_{CV} and n_{eff} alongside AUC.

Minimal algorithm: (1) construct labels only for rows whose H -step outcome is observed; (2) choose the split protocol that matches the deployment

target; (3) fit preprocessing on each training fold only; (4) fit the model and predict held-out rows; (5) aggregate pooled and episode-weighted metrics from aligned out-of-fold predictions; and (6) optionally rerun row-wise KFold and bootstrap episodes to quantify Δ_{CV} uncertainty.

Specificity and failure modes: a large positive Δ_{CV} is a warning flag, not a proof of episode memorization. Distribution shift between grouped folds, severe fold imbalance, episode heterogeneity, or a mismatch between row-wise interpolation and the true deployment target can also enlarge the gap. Near-zero gaps can likewise be false negatives when dependence is weak or model capacity is limited, as illustrated by the logistic-regression cells in Table 4. Practical warning signs are therefore: large positive Δ_{CV} ; substantial disagreement between row-wise and grouped CV; and unusually strong discrimination despite a very small number of episodes.

8 Discussion

The main lesson is not that pseudoreplication is always larger than every other leakage mechanism. Rather, row-wise CV can by itself create severe optimism when expanded rows retain strong episode-specific dependence. This matters because row-wise splitters are convenient defaults in general-purpose ML libraries and can be applied mechanically to expanded panels even when they are inappropriate for the underlying prediction target.

Row-wise validation is not always invalid. If the deployment target is interpolation within already observed episodes—for example, forecasting future rows for machines, patients, or regions that are already represented in the training history—then a row-wise or temporally blocked within-episode split may estimate a relevant risk. Our critique applies when the target is generalization to new episodes, where the independent sampling unit is the episode rather than the expanded row.

A chronological, temporally blocked split within each episode is a common intuition for “safe” validation, since it never trains on rows chronologically after a test row – but it protects against only one of the three failure modes in Table 1. In our main benchmark, temporal-blocked within-episode validation reaches $AUC = 0.917$ (Section 6), far closer to the row-wise memorization figure of 0.835 than to the grouped, unseen-episode figure of 0.552. This is expected: temporal blocking still trains on other rows from the *same* episode, letting the model condition on episode identity and history, which unseen-episode generalization rules out. A temporal split is right when the target really is forecasting future rows of already-observed units, but it is not a substitute for episode-grouped validation when the target is new-episode generalization, and “we used a time-based split” is not by itself sufficient evidence against pseudoreplication.

Table 7 turns the paper’s warning into a practical audit aid for panel-expanded forecasting studies.

Our study has limitations. All evidence is simulation-based: we do not include real-data reanalyses or sequence models, and we do not attempt to estab-

Table 7. Audit checklist for panel-expanded event forecasting studies.

Question	Why it matters
What is the independent episode unit?	Defines the generalization target and the appropriate grouping variable.
What is the deployment target: new-episode generalization or forecasting later rows of known episodes?	Determines whether grouped, temporal, or row-wise validation is appropriate; a temporal split alone does not certify unseen-episode generalization.
Are rows expanded from episodes?	Determines whether row-wise splitting can create dependence overlap.
Can rows from the same episode appear in both train and test folds?	Directly tests for pseudoreplication.
Are preprocessing parameters fit within each training fold?	Detects normalization, imputation, or filtering leakage.
Are both row-wise and grouped estimates reported?	Makes Δ_{CV} observable rather than implicit.

lish how prevalent this problem is in published practice. Although the robustness grid now spans linear, bagged-tree, and boosted-tree models with ten replicates per cell, and the higher-signal configuration in Section 6.1 shows the gap survives a genuinely non-trivial grouped baseline, a controlled synthetic DGP cannot capture every source of episode heterogeneity present in real panels. The normalization-leakage evidence used full-trajectory episode-wise normalization on a drift DGP; simple global standardization alone had no measurable effect, and even at 30 replicates per model family the episode-wise effect is modest and, for two of three model families, only borderline distinguishable from seed-to-seed noise (Section 6). Because the label asks whether an event occurs within the next 14 days, the simulated row-level positive prevalence is not ultra-rare despite the low underlying daily hazard.

9 Conclusions

This paper studied a mechanistically explicable way in which a widely used default – row-wise K-fold cross-validation applied to panel-expanded event data – produces misleadingly optimistic results, and proposed a diagnostic that costs nothing beyond re-running an existing evaluation with a different splitter. Relative to the taxonomies and domain-specific audits in Section 2, its contribution is a controlled data-generating process isolating pseudoreplication’s effect size from explicit and preprocessing leakage under one shared protocol, stress-tested across dependence, episode counts, and model families.

Three findings summarize the empirical contribution. Row-wise KFold increases AUC by 0.283 (from 0.552 to 0.835) in the main benchmark, comparable to a deliberate time-to-event leak (+0.261), and the gap remains large and significant ($\Delta_{CV} = 0.199$) even once the grouped baseline itself reaches a

genuinely non-trivial AUC of 0.723. The gap grows with model capacity and within-episode dependence, is largest for random forests and boosted trees, and is nonzero even at zero serial autocorrelation – confirming that latent episode heterogeneity and shared event-time structure, not autocorrelation alone, drive the effect. And preprocessing leakage from full-trajectory episode normalization consistently, though modestly, inflates grouped-CV AUC under a plausible pre-event drift pattern (+0.034 to +0.063 across model families at 30 replicates each), clearly distinguishable from noise only for random forests – underscoring that magnitude claims about leakage effects need substantially more replication than the main pseudoreplication result does, a caution we apply to our own numbers throughout. Panel expansion creates evaluative value only if validation respects episode boundaries; reporting grouped alongside row-wise CV, fitting preprocessing within each training fold, reporting Δ_{CV} and n_{eff} , and working through the audit checklist in Table 7 make that optimism easy to detect at negligible added cost.

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